

DESIGN & SYSTEMS DOCUMENT

Dealflow Platform

Venture-capital deal-flow & portfolio-management system —
architecture, external services, and data flows.

Next.js 16

React 19

Vercel

Supabase

Google Workspace APIs

Google Gemini

Nodemailer

Recharts

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1. What the platform is

An internal venture-capital deal-flow & portfolio-management platform for a fund — a single Next.js web application that gives the investment team one place to run the entire investment lifecycle.

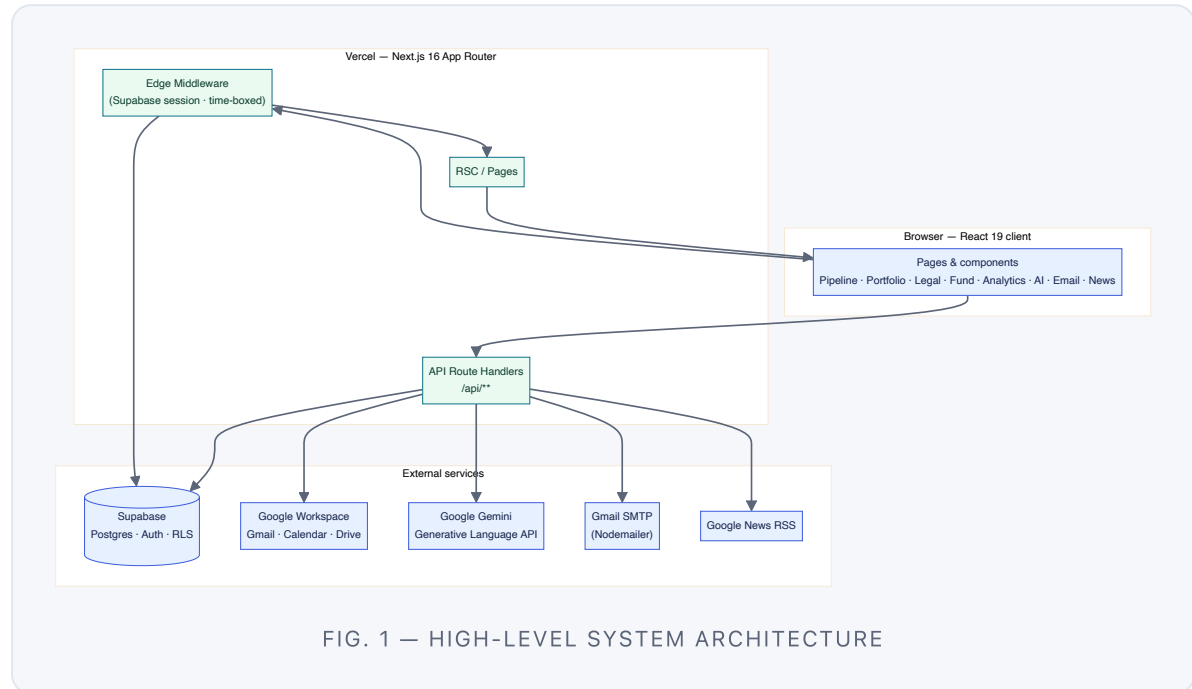
- **Deal pipeline** — track inbound startups (Kanban + table).
- **Portfolio** — rounds, ownership, health, KPIs.
- **Legal** — documents & rights matrix.
- **Fund** — capital, bank statements, allocation.
- **Analytics** — pipeline conversion & fund performance.
- **AI** — deck analysis, call/meeting analysis, KPI extraction, IC memos.
- **Email / Calendar / News** — Gmail, scheduling, company news.
- **Audit & Admin** — activity log, roles, invites.

2. Tech stack at a glance

Layer	Technology
Framework	Next.js 16 (App Router, React 19, TypeScript)
Hosting / runtime	Vercel — serverless API routes + edge middleware
Database + Auth + Storage	Supabase (Postgres, Auth, Row-Level Security)
AI / LLM	Google Gemini (Generative Language API, multimodal)
Email — invites	Gmail SMTP via Nodemailer ; Resend SDK available
Email / Calendar / Drive	Google Workspace APIs via <code>googleapis</code> + OAuth 2.0
News	Google News RSS
Charts / DnD	Recharts / @hello-pangea/dnd
Doc export	docx , jsPDF , xlsx , file-saver
Icons / Dates	lucide-react / date-fns

3. System architecture

Everything runs as one Next.js app on Vercel. The browser talks to App Router pages and API route handlers; route handlers fan out to the external services below.



4. External services (what we depend on)

4.1 Vercel — hosting & delivery

Deployment platform for the Next.js app. Runs App Router pages, **API route handlers** (`src/app/api/**`) and **edge middleware** (`src/middleware.ts`). Middleware was hardened to **time-box Supabase calls to avoid 504 gateway timeouts** under Vercel's serverless function limits.

4.2 Supabase — database, auth, storage

- **Postgres** — all application data; schema in `supabase/*.sql` .
- **Auth** — sign-in & sessions via `@supabase/ssr` .
- **Row-Level Security** — per-row policies (`fix-rls-recursion.sql` , `permissions-redesign.sql`).
- Three clients: `client.ts` (browser, anon key), `server.ts` (route handlers), `middleware.ts` (session refresh). Server uses the **service-role key** for admin/invites.

4.3 Google Workspace — OAuth + Gmail + Calendar + Drive

Per-user OAuth 2.0 via `googleapis` (`src/lib/google.ts`). Scopes: `gmail.send` , `gmail.readonly` , `calendar` , `calendar.events` , `drive.readonly` . Tokens persist in the `google_tokens` table. Powers Gmail send/read/ingest, Calendar events & booking links, and read-only Drive deck pulls.

4.4 Google Gemini — AI / LLM

Shared multimodal client (`src/lib/gemini.ts`) calling `generativelanguage.googleapis.com` , with a model fallback chain and a backup API key for quota resilience. Powers `deck-analysis` , `call-analysis` , `meeting-analysis` , `kpi-extraction` , `ic-memo` , `rejection-email` , `quick-summary` , `filter-brief` .

4.5 Email — Nodemailer (Gmail SMTP) & Resend

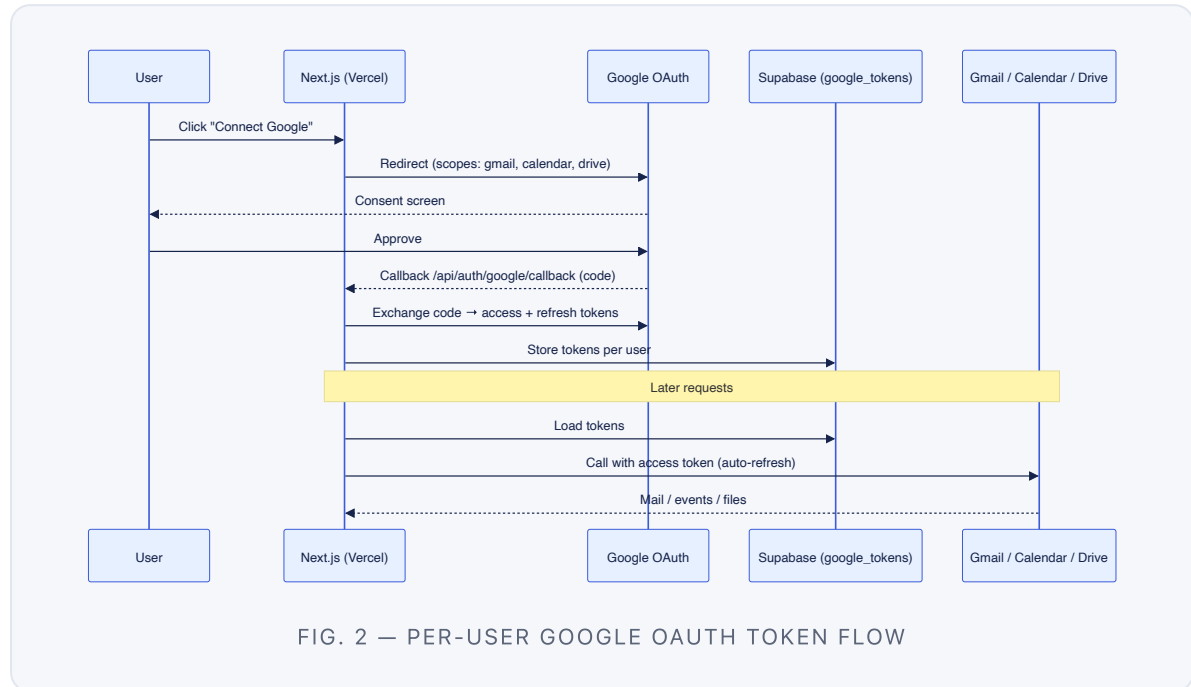
Nodemailer over **Gmail SMTP** (`GMAIL_USER` + `GMAIL_APP_PASSWORD`) sends **user-invite emails** — distinct from the per-user Gmail API. **Resend** SDK is installed as an alternative provider.

4.6 Google News RSS & client libraries

The News page fetches **Google News RSS**, parses XML, and auto-categorizes + scores sentiment. Client packages: `Recharts` (charts), `@hello-pangea/dnd` (Kanban), `docx/jsPDF/xlsx/file-saver` (exports), `lucide-react`, `date-fns`.

5. Google OAuth & data flow

Each user links their own Google account; refresh tokens are stored server-side and reused for Gmail/Calendar/Drive calls.



6. AI request & fallback flow

The Gemini client retries across models and keys so a single exhausted quota pool doesn't break AI features.

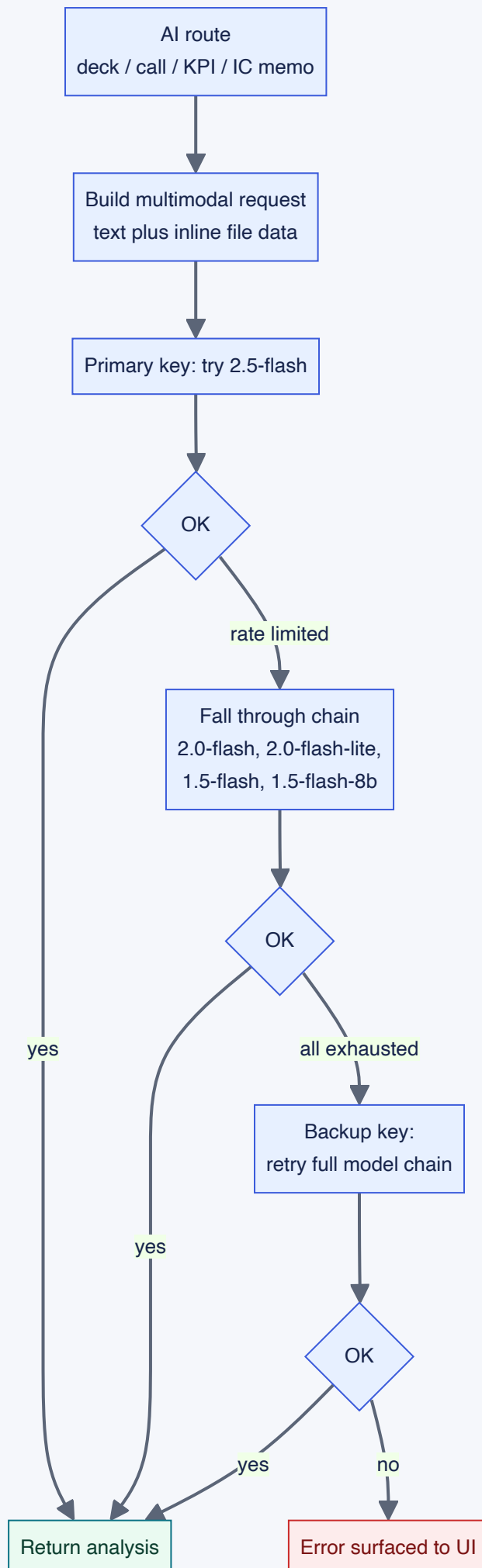
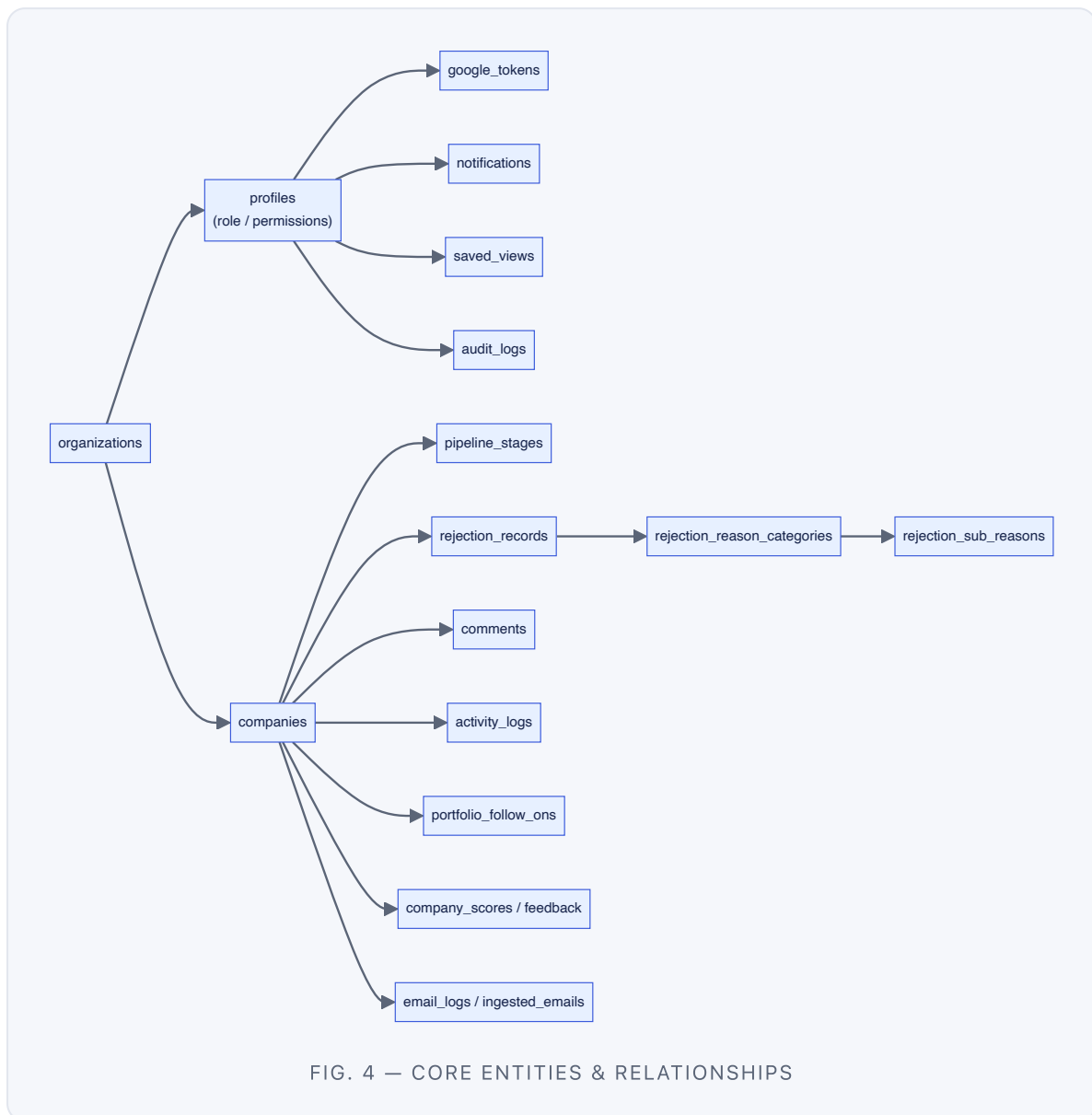




FIG. 3 — GEMINI MODEL + API-KEY FALLBACK CHAIN

7. Data model (Supabase)



Other tables: `industries`, `deal_source_names`, `pending_invites`, share-count & entry-stage extensions, stage deadlines, booking tokens.

8. Authentication & authorization

#	Mechanism	Purpose
1	Supabase Auth	App login; sessions refreshed by middleware
2	Role permissions + RLS	Gate admin features; enforce row access in Postgres
3	Google OAuth 2.0	Per-user Gmail/Calendar/Drive grant; tokens in <code>google_tokens</code>
4	Invites (Nodemailer)	Admins invite users via <code>pending_invites</code>
5	Audit trail	<code>audit_logs</code> + <code>activity_logs</code> record actions

9. Environment configuration

Variable	Service	Purpose
<code>NEXT_PUBLIC_SUPABASE_URL</code>	Supabase	Project URL
<code>NEXT_PUBLIC_SUPABASE_ANON_KEY</code>	Supabase	Public anon key (RLS-enforced)
<code>SUPABASE_SERVICE_ROLE_KEY</code>	Supabase	Privileged server key (admin/invites)
<code>GOOGLE_CLIENT_ID</code> / <code>_SECRET</code>	Google OAuth	OAuth client credentials
<code>GOOGLE_REDIRECT_URI</code>	Google OAuth	OAuth callback URL
<code>GMAIL_USER</code> / <code>GMAIL_APP_PASSWORD</code>	Gmail SMTP	Nodemailer invite sender
<code>GEMINI_API_KEY</code> (+ <code>_BACKUP</code>)	Google Gemini	LLM keys (primary + fallback)

10. Service responsibility & failure impact

Service	Owns	If it fails
Vercel	Hosting, functions, middleware	Whole app down
Supabase	All data, auth, RLS	App unusable (no data/login)
Google Workspace	Gmail, Calendar, Drive (per user)	Email/calendar/deck features fail; core app fine
Google Gemini	All AI analysis	AI fails; mitigated by model + key fallback
Gmail SMTP	Invite emails	Can't invite new users
Resend	(Available) transactional email	N/A unless adopted
Google News RSS	Company news	News page empty

11. Notes & considerations

Google concentration: Supabase + Google (OAuth, Gmail, Calendar, Drive, Gemini, News) cover nearly all external dependencies — single-ecosystem risk is high.

- **Quota resilience** exists for Gemini (model chain + backup key) but not for the Workspace APIs.
- **Serverless timeouts:** middleware Supabase calls are time-boxed for Vercel — keep blocking middleware calls lean.
- **Two email paths** coexist: app-level Nodemailer invites vs. per-user Gmail API — don't conflate them.